PERSONAL EXPENSE CATEGORY PREDICTOR

Mini Project Report

Technology: Python

Project Type: Beginner-friendly rule-based expense categorization

Prepared for learning, demonstration, and GitHub portfolio use.

1. Abstract

The Personal Expense Category Predictor is a simple Python application that categorizes an expense based on a keyword entered by the user. It helps demonstrate how conditional statements, functions, string handling, and user input can be used to organize everyday expenses. This version uses predefined rules; it does not train or use a machine-learning model.

2. Problem Statement

People often record expenses using short descriptions such as groceries, petrol, or notebooks. Manually sorting these descriptions into categories can be repetitive. This mini project provides a basic automated approach for assigning a category to a known keyword.

3. Objectives

• Accept an expense keyword from the user.

• Convert the input to lowercase to make matching less sensitive to capitalization.

• Assign a category using predefined rules.

• Display the predicted category clearly.

• Provide a simple starting point for future improvements.

4. Software Requirements

Requirement	Details
Programming language	Python 3
Editor	VS Code, IDLE, or another Python editor
Libraries	No external libraries required
Input	Expense keyword typed by the user
Output	Predicted expense category

5. Algorithm

1. Start the program.

2. Define a function that accepts an expense keyword.

3. Convert the keyword to lowercase.

4. Compare the keyword with the predefined lists.

5. Return Food, Transport, Shopping, Education, or Other.

6. Display the result and end the program.

6. Source Code

```
# Personal Expense Category Predictor

def predict_category(expense):
    expense = expense.lower().strip()

    if expense in ["food", "restaurant", "groceries"]:
        return "Food"
    elif expense in ["bus", "petrol", "train"]:
        return "Transport"
    elif expense in ["shirt", "shoes", "dress"]:
        return "Shopping"
    elif expense in ["book", "pen", "notebook"]:
        return "Education"
    else:
        return "Other"


print("Personal Expense Category Predictor")
print("-----------------------------------")

expense = input("Enter your expense: ")
category = predict_category(expense)

print("Predicted Category:", category)
```

7. Sample Output

```
Personal Expense Category Predictor
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Enter your expense: groceries
Predicted Category: Food
```

8. Testing

Input	Expected category
groceries	Food
petrol	Transport
shoes	Shopping
notebook	Education
movie	Other

9. Advantages and Limitations

Advantages: Simple to understand, quick to run, and needs no external libraries.

Limitations: Only predefined keywords are recognized. Unlisted descriptions are categorized as Other. It is not a machine-learning model and does not track amounts or create charts.

10. Future Enhancements

• Add more keywords and allow users to maintain their own category lists.

• Record expense amount, date, and description in a CSV file.

• Create monthly summaries and charts.

• Build a simple web interface.

• As a separate advanced version, train and evaluate a machine-learning classifier using labeled expense descriptions.

11. Conclusion

This mini project demonstrates a basic approach to organizing expense descriptions with Python. It provides a foundation for learning functions, conditions, input/output, and project documentation. The rule-based version can be extended gradually into a more complete expense management application.